John Glen Siy

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Education

University of California, Berkeley — **Berkeley, CA**
B.S. Electrical Engineering and Computer Science — Graduated December 2025
Relevant Coursework:
Data Structures, Computer Architecture, Advanced Algorithms, Computer Security, Artificial Intelligence, Machine Learning, Database Systems, Optimization Models, Reinforcement Learning, Deep Learning, UI Design and Development

Experience

Freelance Work — **December 2025 – Present**
Full Stack Engineer and Web Development Consultant
- Developed a biomechanics web app with React, TypeScript and FastAPI, helping local tennis players and coaches improve their form
- Designed asynchronous video processing with Celery and Redis, keeping the app fast and responsive while videos process
- Built a marketing website for an Airbnb host, integrating Hotjar heatmap and click-tracking analytics to identify visitor drop-off points and guide UX improvements, increasing monthly booking inquiries by 2 to ~10

Codeology @ UC Berkeley — **February 2023 – December 2025**
Full Stack Engineer and Product Manager
- Built a full-stack language translation web app with React, TypeScript and HuggingFace, leading a team of 5 junior developers and contributing to over $10,000 in profits for the organization
- Developed an ML pipeline with TensorFlow and Keras to classify audio clips into 20+ music genres, achieving 94% test set accuracy
- Wrote Python scripts to optimize SEO term selection, boosting technical explainers to top position for relevant search queries

CozmicGo — **June 2024 – August 2024**
Software Engineer Contractor
- Migrated a cross-platform event planning mobile app from Swift to TypeScript and React Native
- Implemented geospatial query pagination with Firebase to optimize user feeds, eliminating all critical latency issues for event filters
- Facilitated A/B testing for different UIs, increasing average user session length by 85% and decreasing monthly churn rate by 40%

Projects

"Real" Browser Extension | GitHub | Website
- Built a Firefox extension inspired by the sports app Real that overlays detailed real-time sports results onto Google search pages, eliminating the need to click through separate stat sites to check scores (React/Redux/TypeScript/MongoDB/Express)
- Configured Playwright web scrapers on Fly.io containers, publishing live events over Redis pub/sub
- Designed a horizontally scaling SSE delivery tier to automatically provision Express instances as concurrent viewers grow, capable of supporting 1,000+ simultaneous connections

RallyGrids
- Daily tennis trivia game built with React, TypeScript and Postgres, inspired by HoopGrids
- Wrote a SQL migration script to normalize and backfill a public ATP/WTA dataset into Postgres, using a versioned category registry to keep puzzle generation forward compatible
- Automated daily puzzle deployment with a GitHub Actions job that generates, validates and caches puzzles ahead of release

Raft Consensus Protocol *(MIT 6.5840)*
- Implemented the Raft consensus protocol in Go for fault-tolerant replicated state servers with leader election, heartbeats, log replication and commit advancement under a simulated RPC network that drops, delays and reorders messages
- Tested successfully against a 5s re-election upper bound with 400–800ms election timeouts, achieving leader recovery in ~1.2s

Tennis Match Analysis
- Built a computer vision pipeline that detects players and the ball via YOLOv8 and extracts court keypoints with a CNN, outputting per-player speed, shot speed and shot counts overlaid on the source video
- Extended the pipeline to infer rally segments by detecting serve initiations from ball-toss trajectory and player baseline position

Gossip Glomers Distributed Systems Challenge | GitHub
- Solved a series of distributed systems problems in Go, including gossip broadcast protocols and grow-only CRDT counters
- Culminated in building a multi-node transactional key-value store with a Kafka-style commit log and Read Committed isolation

SQL Query Optimizer | GitHub
- Implemented join algorithms (block nested loop, grace hash, sort-merge) for a query engine, including memory-efficient partitioning to handle joins too large to fit in memory
- Built a cost-based query optimizer, estimating I/O costs to choose join order and decide between index vs. table scans

Secure File Sharing System | GitHub
- Designed an encrypted file sharing system in Go with secure user login, appending file contents, and sharing and revoking access
- Optimized write cost to O(append size) by storing each append at a UUID derived from file location and append count, avoiding a full re-encryption of the file on every edit

Voice-Controlled Car | Design Doc
- Built a voice-controlled autonomous car with a mic-board front-end and op-amp bandpass filter, control firmare in C on a TI MSP430
- Characterized each wheel's PWM-to-velocity response via least-squares regression and implemented a closed-loop controller with a feedback term on encoder delta
- Classified 4 spoken commands using PCA for dimensionality reduction and k-means on the projected features

Technical Skills

Languages: Python, TypeScript, JavaScript, Go, Java, C, C++, SQL
Frontend: React, Next.js, React Native, HTML/CSS, Tailwind, Redux, Jest
Backend: Node.js, REST API, FastAPI, Firebase, PostgreSQL, MongoDB, GraphQL, MySQL, AWS, Docker, Redis
Tools & Libraries: Git, GitHub Actions, Cursor, Claude Code, Claude Design, Terraform, Postman